

Moveit Inverse Kinematics Design

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- 1. Environment
- 2. Driving the Real Machine
 - 2.1 Starting the Real Machine
 - 2.2 Starting MoveIt2
- 3. Inverse Kinematics Design
 - Startup Command
 - Notes

1. Environment

Ubuntu 22.04

ROS2: Humble

2. Driving the Real Machine

Driving the real machine involves subscribing to MoveIt2's `/joint_states` topic to convert the joint state information of the robotic arm into control data for the actual robotic arm.

Note: Since the real robotic arm lacks obstacle avoidance, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid locations with obstacles.

(It is recommended to use preset poses for the demonstration of driving the real machine)

2.1 Starting the Real Machine

If not driving the real machine, demonstrate the robotic arm movements in MoveIt simulation (skip the following steps):

```
ros2 run dofbot_driver dofbot_driver
```

2.2 Starting MoveIt2

```
ros2 launch dofbot_moveit demo.launch.py
```

3. Inverse Kinematics Design

Inverse kinematics refers to the process where, given the position of the robotic arm's end effector, MoveIt calculates the angles of each joint of the robotic arm and then automatically plans the target pose.

Startup Command

After the robotic arm has successfully loaded MoveIt and displays `You can start planning now!`, run the following command: the robotic arm will automatically plan its position.

```
ros2 run dofbot_moveit set_target_position
```

Notes

Since the position of the robotic arm's end effector is difficult to determine, it is recommended that users find a reachable pose using the following method.

Check the pose of `Arm5_Link` and then replace the target pose in the program: data changes will only occur after the robotic arm is planned and executed to the specified pose.

MotionPlanning → Planned Path → Links: Positions, Orientation



